

Wine Quality Prediction

Classification of Wine Quality Based on Chemical Properties

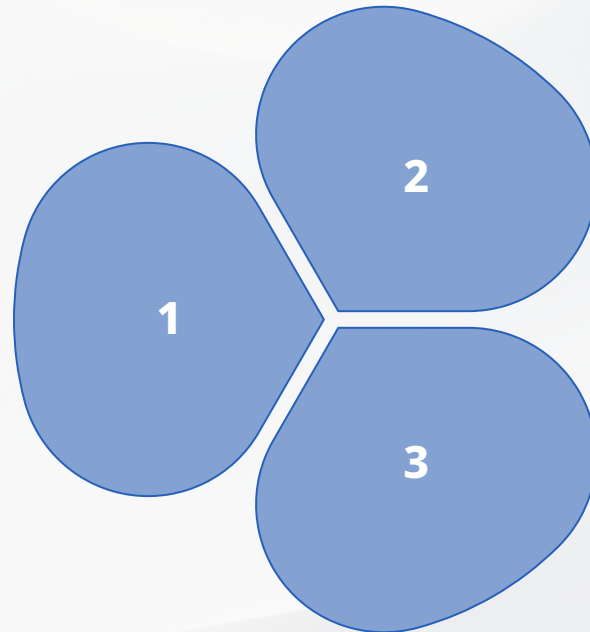
Tools Used: Python, Pandas, NumPy, Scikit-learn, Seaborn, Matplotlib

PROJECT OBJECTIVE

Problem Statement

The Challenge

Wine quality is traditionally judged by experts — subjective and costly. This project replaces that with data-driven predictions.



The Approach

Use laboratory chemical measurements as input features to train machine learning classification models.

The Goal

Train and compare multiple classification models to identify the best performing algorithm for wine quality prediction.

Wine Quality Dataset Overview

Feature	Description
Alcohol	Alcohol percentage
pH	Acidity level
Density	Liquid density
Sulphates	Preservative level
Volatile Acidity	Vinegar acidity
Quality	Target Variable

説明変数→	alcohol	malic_acid	ash	acidity of ash	magnesium	total_phenols	flavanoids	nonflavanoid phenols	proanthocyanins	color_intensity	hue	od280/od315 of diluted wines	proline	target
行番号↓	アルコール度数	リンゴ酸	灰分(かいふん)	灰分のアルカリ度	マグネシウム	全フェノール含量	フラボノイド	非フラボノイドフェノール	プロアントシアニン	色の濃さ	色相	希釈ワイン溶液のOD280/OD315	プロリン	ワインの品質
0	14.23	1.71	2.43	15.60	127	2.80	3.05	0.28	2.29	5.64	1.04	3.92	1065	0
1	13.20	1.78	2.14	11.20	100	2.65	2.76	0.26	1.28	4.38	1.05	3.40	1050	0
2	13.16	2.36	2.67	16.60	101	2.80	3.24	0.30	2.81	5.68	1.03	3.17	1185	0
3	14.37	1.95	2.50	16.80	113	3.85	3.49	0.24	2.18	7.80	0.86	3.45	1480	0
4	13.24	2.59	2.87	21.00	118	2.80	2.69	0.39	1.82	4.32	1.04	2.93	735	0
5	14.20	1.76	2.45	15.20	112	3.27	3.39	0.34	1.97	6.75	1.05	2.85	1450	0
6	14.39	1.87	2.45	14.60	96	2.50	2.52	0.30	1.98	5.25	1.02	3.58	1290	0
7	14.06	2.15	2.61	17.60	121	2.60	2.51	0.31	1.25	5.05	1.06	3.58	1295	0
8	14.83	1.64	2.17	14.00	97	2.80	2.98	0.29	1.98	5.20	1.08	2.85	1045	0
9	13.86	1.35	2.27	16.00	98	2.98	3.15	0.22	1.85	7.22	1.01	3.55	1045	0
10	14.10	2.16	2.30	18.00	105	2.95	3.32	0.22	2.38	5.75	1.25	3.17	1510	0
59	12.37	0.94	1.36	10.60	88	1.98	0.57	0.28	0.42	1.95	1.05	1.82	520	1
60	12.33	1.10	2.28	16.00	101	2.05	1.09	0.63	0.41	3.27	1.25	1.67	680	1
61	12.64	1.36	2.02	16.80	100	2.02	1.41	0.53	0.62	5.75	0.98	1.59	450	1
62	13.67	1.25	1.92	18.00	94	2.10	1.79	0.32	0.73	3.80	1.23	2.46	630	1
63	12.37	1.13	2.16	19.00	87	3.50	3.10	0.19	1.87	4.45	1.22	2.87	420	1
64	12.17	1.45	2.53	19.00	104	1.89	1.75	0.45	1.03	2.95	1.45	2.23	355	1
65	12.37	1.21	2.56	18.10	98	2.42	2.65	0.37	2.08	4.60	1.19	2.30	678	1
66	13.11	1.01	1.70	15.00	78	2.98	3.18	0.26	2.28	5.30	1.12	3.18	502	1
67	12.37	1.17	1.92	19.60	78	2.11	2.00	0.27	1.04	4.68	1.12	3.48	510	1
68	13.34	0.94	2.36	17.00	110	2.53	1.30	0.55	0.42	3.17	1.02	1.93	750	1
131	12.88	2.99	2.40	20.00	104	1.30	1.22	0.24	0.83	5.40	0.74	1.42	530	2
132	12.81	2.31	2.40	24.00	98	1.15	1.09	0.27	0.83	5.70	0.66	1.36	560	2
133	12.70	3.55	2.36	21.50	106	1.70	1.20	0.17	0.84	5.00	0.78	1.29	600	2
134	12.51	1.24	2.25	17.50	85	2.00	0.58	0.60	1.25	5.45	0.75	1.51	650	2
135	12.60	2.46	2.20	18.50	94	1.62	0.66	0.63	0.94	7.10	0.73	1.58	695	2
136	12.25	4.72	2.54	21.00	89	1.38	0.47	0.53	0.80	3.85	0.75	1.27	720	2
137	12.53	5.51	2.64	25.00	96	1.79	0.60	0.63	1.10	5.00	0.82	1.69	515	2
176	13.17	2.59	2.37	20.00	120	1.65	0.68	0.53	1.46	9.30	0.60	1.62	840	2
177	14.13	4.10	2.74	24.50	96	2.05	0.76	0.56	1.35	9.20	0.61	1.60	560	2

Initial Inspection

- Checked shape and data types
- Verified missing values
- Examined quality score distribution

EDA

Distribution of Chemical Features



Alcohol

Follows a near-normal distribution across samples



Volatile Acidity

Contains notable outliers requiring attention



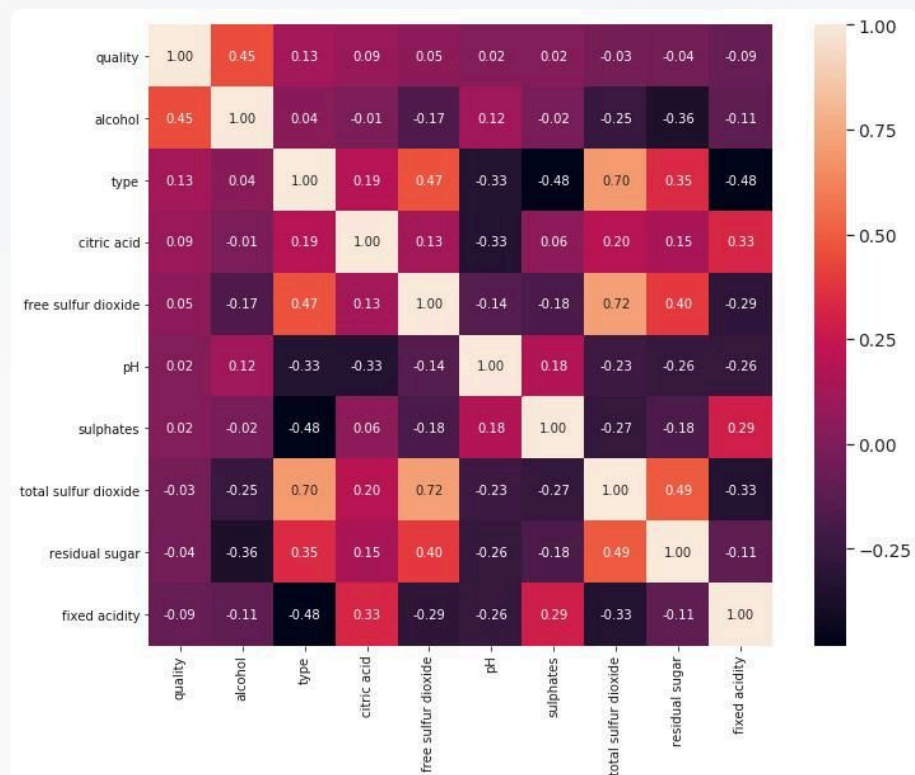
Sulphates

Vary considerably across wine samples

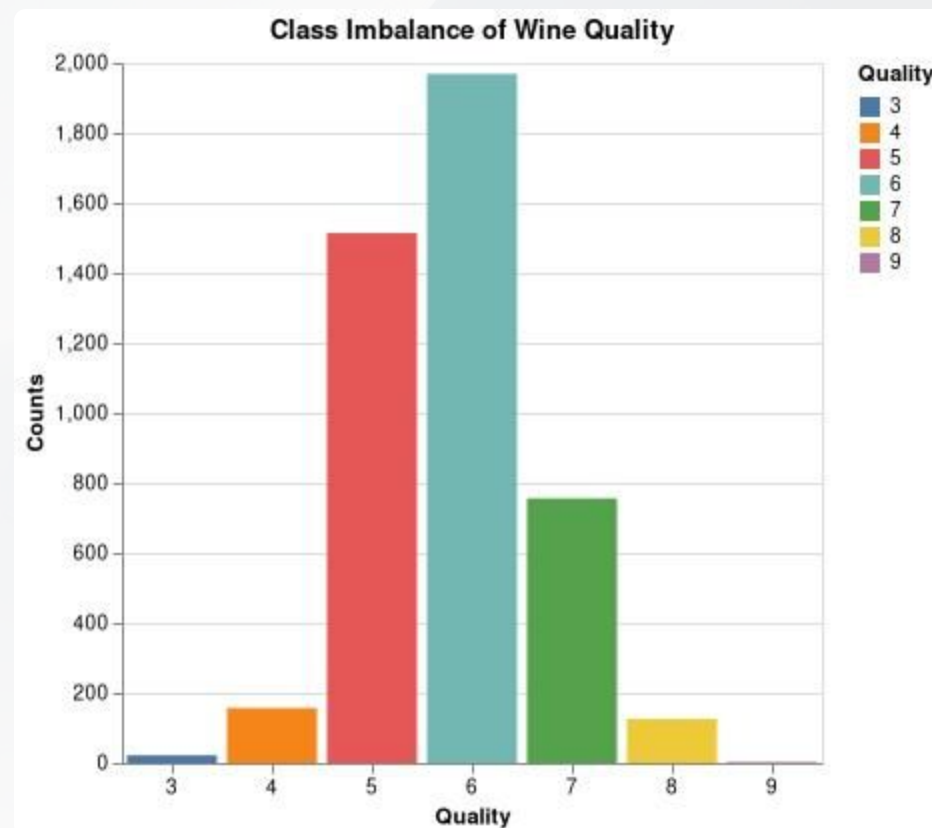
Insight: Understanding feature distributions improves preprocessing and model quality.

Correlation & Class Distribution

Correlation Heatmap



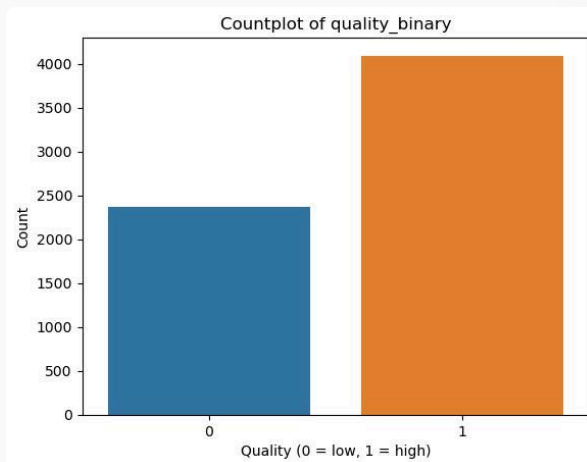
Quality Score Distribution



Quality scores 5 and 6 dominate the dataset, creating class imbalance.

Converting Quality into Binary Classes

Original Score	Class
3–5	Bad Wine (0)
6–8	Good Wine (1)



Why Binary Classification?

01

Reduce Class Imbalance

Merges minority classes to create a more balanced dataset

02

Improve Accuracy

Fewer target classes yield better model precision

03

Business Clarity

Easier to interpret: simply good or bad wine

MODELLING

Machine Learning Models

Train/Test Split: 80% Training • 20% Testing • Stratified Sampling



Random Forest

Ensemble classifier combining multiple decision trees for robust predictions



SGD Classifier

Stochastic gradient descent linear classifier optimised for large datasets

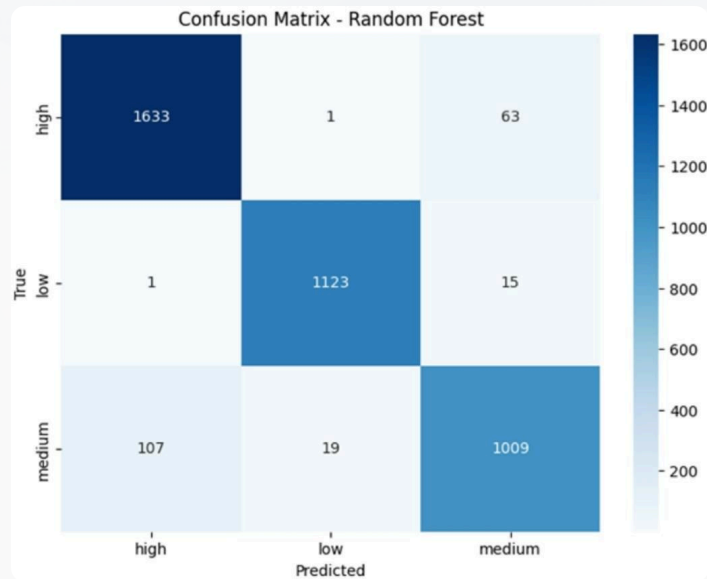


Support Vector Classifier

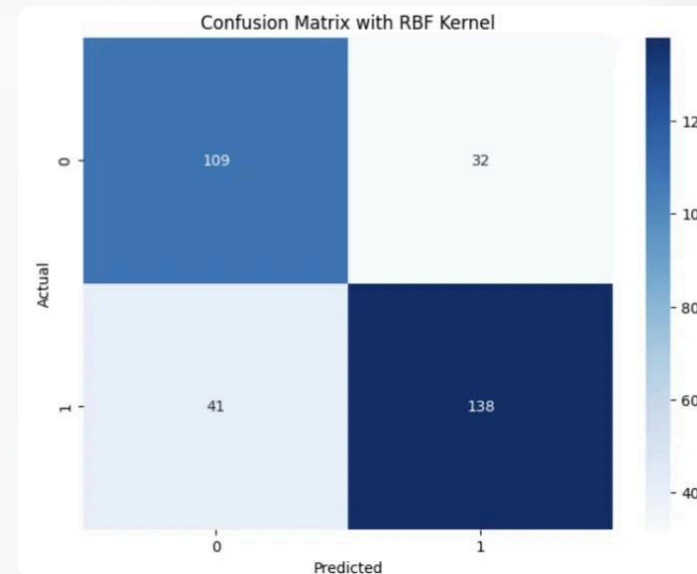
Margin-based classifier that finds optimal decision boundaries between classes

Model Evaluation

Confusion Matrix — Random Forest



Confusion Matrix — SVC (RBF Kernel)



Classification Metrics

Accuracy — Overall model performance

Precision — Correct positive predictions

Recall — Detects actual positives

F1 Score — Balanced evaluation metric

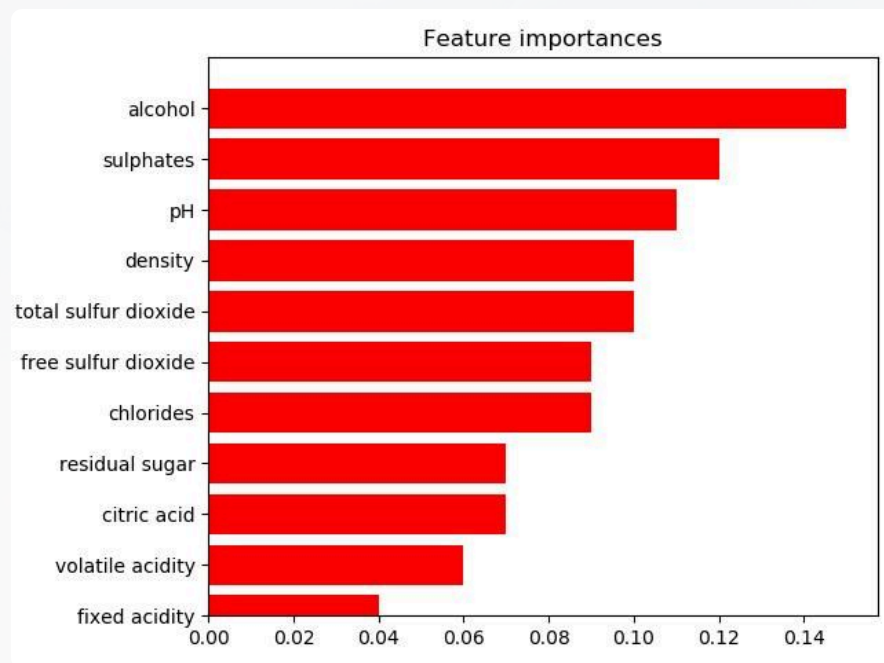
Observation: Random Forest achieved the strongest overall performance across all metrics.

Z

RESULTS

Feature Importance & Model Comparison

Random Forest Feature Importance



Top Features

1 Alcohol

Highest predictive importance

2 Volatile Acidity

Strong negative correlation with quality

3 Sulphates

Key preservative linked to quality

Model Comparison

Model	Performance
Random Forest	Best
SVC	Good
SGD	Moderate

Conclusion & Business Recommendations

Best Model: Random Forest



Why Random Forest?

Highest accuracy, handles nonlinear relationships, and provides feature importance insights.



Focus on Key Features

Direct quality control efforts toward alcohol content and volatile acidity.



Automate Assessment

Deploy Random Forest for automated, objective wine quality assessment.



Continuous Retraining

Collect more samples for underrepresented classes and retrain with new production data.

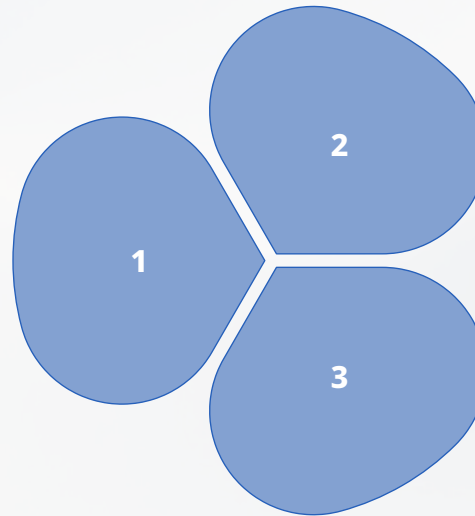
This project successfully demonstrated end-to-end classification, feature engineering, model comparison, and deployment-ready business insights.

PROJECT OBJECTIVE

Business Problem & Goal

Business Problem

Banks process millions of digital transactions every day. Fraudulent transactions are rare but cause significant financial losses if not detected quickly.



Objective

Build a machine learning model that accurately classifies transactions as **Genuine** or **Fraudulent**.

Expected Outcome

A reliable fraud detection system with high **Recall** and strong overall performance.

